

THE THIRTEENTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 23, 1953

Morning Session

1. Prove that, for every positive integer n ,

$$\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}$$

is more than $\frac{2}{3} n \sqrt{n}$ and less than

$$\frac{4n + 3}{6} \sqrt{n}. \quad (\text{page 364})$$

2. Six points are in general position in space (no three in a line, no four in a plane). The fifteen line segments joining them in pairs are drawn and then painted, some segments red, some blue. Prove that some triangle has all its sides the same color. (page 365)

3. If x_1, x_2, x_3 are real numbers and the sum of any two is greater than the third, show that

$$\frac{2}{3} \sum_{i=1}^3 x_i \sum_{i=1}^3 x_i^2 > \sum_{i=1}^3 x_i^3 + x_1 x_2 x_3. \quad (\text{page 366})$$

4. From the identity

$$\begin{aligned} \int_0^{\pi/2} \log \sin 2x \, dx &= \int_0^{\pi/2} \log \sin x \, dx \\ &+ \int_0^{\pi/2} \log \cos x \, dx + \int_0^{\pi/2} \log 2 \, dx, \end{aligned}$$

deduce the value of

$$\int_0^{\pi/2} \log \sin x \, dx. \quad (\text{page 368})$$

5. Let P be a point from which three distinct normals can be drawn to a parabola. Show that the sum of the angles which these three normals make

with the axis exceeds by a multiple of π the angle which the line joining P to the focus makes with the axis. (page 369)

6. Show that the sequence

$$\sqrt{7}, \sqrt{7 - \sqrt{7}}, \sqrt{7 - \sqrt{7 + \sqrt{7}}}, \sqrt{7 - \sqrt{7 + \sqrt{7 - \sqrt{7}}}}, \dots$$

converges, and evaluate the limit. (page 370)

7. Assuming that the roots of $x^3 + px^2 + qx + r = 0$ are all real and positive, find the relation between p , q , and r which is a necessary and sufficient condition that the roots may be the cosines of the angles of a triangle. (page 372)

Afternoon Session

1. Is the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n^{(n+1)/n}}$$

convergent? Prove your statement. (page 374)

2. Let $a_0, a_1, \dots, a_n$ be real numbers and let $f(x) = a_0 + a_1x + \dots + a_nx^n$. Suppose that, for every integer i , $f(i)$ is an integer. Prove that $n! \cdot a_k$ is an integer for each k . (page 374)

3. Solve the equations

$$\frac{dy}{dx} = z(y + z)^n \quad \frac{dz}{dx} = y(y + z)^n,$$

given the initial conditions $y = 1$ and $z = 0$ when $x = 0$. (page 376)

4. Determine the equation of a surface in three dimensional cartesian space which has the following properties: (a) it passes through the point $(1, 1, 1)$; and (b) if the tangent plane be drawn at any point P , and A , B and C are the intersections of this plane with the x , y and z axes respectively, then P is the orthocenter (intersection of the altitudes) of the triangle ABC . (page 377)

5. Show that the roots of $x^4 + ax^3 + bx^2 + cx + d = 0$, if suitably numbered, satisfy the relation $r_1/r_2 = r_3/r_4$, provided $a^2d = c^2 \neq 0$. (page 378)

6. P and Q are any points inside a circle (C) with center C , such that $CP = CQ$. Determine the location of a point Z on (C) such that $PZ = QZ$ shall be a minimum. (page 379)

Note. This is the text of the problem as it was presented to the contestants. The equals sign in the last sentence is, of course, a misprint. It should read: "Determine the location of a point Z on (C) such that $PZ + QZ$ shall be a minimum."

7. Let w be an irrational number with $0 < w < 1$. Prove that w has a unique convergent expansion of the form

$$w = \frac{1}{p_0} - \frac{1}{p_0 p_1} + \frac{1}{p_0 p_1 p_2} - \frac{1}{p_0 p_1 p_2 p_3} + \dots,$$

where $p_0, p_1, p_2, \dots$ are integers and $1 \leq p_0 < p_1 < p_2 < \dots$. If $w = \frac{1}{2}\sqrt{2}$, find p_0, p_1, p_2 . (page 384)